

SUNSPOKE · DESIGN DECISION RECORD

SunSpoke TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sunspoke		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on the TRL 2 review and the SwapCell interface v0.3 items

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 8 and 10 to 13); items 14 to 18 remain proposed, awaiting Amish

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-24) listed nine items as "Proposed, awaiting Amish", and SSP-PRC-001 v0.2 marked every key design choice the same way. On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." He also approved three cross-cutting additions to the SwapCell interface and a portfolio rule for pricing shared packs.

This record lists what that instruction decides and what it leaves open because there was no recommendation to accept. TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are in docs/REVIEW.md (2026-09-24) and SSP-PRC-001 v0.2. They are not repeated here.

■ Decision

Table 1. Decided items.

#	Item	Decision	Where it now lives
1	System voltage and pack	Decided by Amish, 2026-09-25: go with recommendation. Option A: 48 V motor and controller on the SwapCell pack. Option B (36 V kit with its own pack) is kept only as a later low-cost variant	SSP-PRC-001 v0.3, SSP-REQ-001 v0.3, README.md
2	Budget	Decided by Amish, 2026-09-25: go with recommendation (a). Keep budget_usd: 250 ; redefine R12 as "bike kit \$250 or less; solar set costed separately"; the pack is excluded	SSP-REQ-001 v0.3 R12, project.yaml unchanged
3	Motor type	Decided by Amish, 2026-09-25: go with recommendation. Geared front hub motor	SSP-PRC-001 v0.3
4	Pack location	Decided by Amish, 2026-09-25: go with recommendation. Pack in the main triangle on the down tube	SSP-PRC-001 v0.3, SSP-DWG-001
5	Wheel build	Decided by Amish, 2026-09-25: go with recommendation. Motor laced locally into a 28 in rim, with a lacing card	SSP-PRC-001 v0.3
6	Assist speed limit	Decided by Amish, 2026-09-25: go with recommendation. 20 km/h default, adjustable only by the mechanic, never above 25 km/h	SSP-REQ-001 v0.3 R1
7	Assist modes	Decided by Amish, 2026-09-25: go with recommendation. Pedal assist plus a 6 km/h walk assist; no throttle	SSP-REQ-001 v0.3 R1
8	Charging	Decided by Amish, 2026-09-25: go with recommendation. Charge on the bike through the cradle charge port, with the host adapter acting as the SwapCell charge host; hub docks remain an option	SSP-PRC-001 v0.3
10	SwapCell interface v0.3 item W (wake)	Decided by Amish, 2026-09-25 (cross-cutting approval). The cradle fits the 10 kΩ INTERLOCK coding resistor	SSP-PRC-001 v0.3, SSP-REQ-001 v0.3 R13
11	SwapCell interface v0.3 item C (charge-discharge mode)	Decided by Amish, 2026-09-25 (cross-cutting approval). The host adapter requests mode 4 when the bike is switched on while charging	SSP-PRC-001 v0.3, R13
12	SwapCell interface v0.3 item V (latch vibration rating)	Decided by Amish, 2026-09-25 (cross-cutting approval). The cradle is a class V1 vehicle receiver with an over-centre preload lever	SSP-PRC-001 v0.3, R13, SSP-DWG-001
13	Shared pack pricing	Decided by Amish, 2026-09-25 (cross-cutting): the SwapCell pack is priced once in the SwapCell repo (\$414, SWC-CAL-001) and excluded from this kit's budget	bom/bom.csv , bom/bom-notes.md , R12

Item 9 of the TRL 2 list (first partner and region) had no recommendation and is item 14 below.

Items that remain open

Table 2. Open items, Proposed, awaiting Amish.

#	Item	Status
14	First co-design partner and region	Proposed, awaiting Amish. No recommendation was made; portfolio rule: partners are chosen per area later
15	Whether light filing of the donor fork slots (about 0.24 mm per side) is acceptable with torque arms fitted (R4)	Proposed, awaiting Amish. No recommendation yet; needs the donor survey
16	Handlebar power switch wired in series with the INTERLOCK coding resistor, and host adapter powered from legacy discharge or the charge inlet (SSP-CAL-001 section 6)	New engineering proposal from this session, awaiting Amish. Recommendation: adopt
17	Motor temperature input on the controller to derate instead of cutting out (R3)	New engineering proposal, awaiting Amish. Recommendation: specify a controller with a thermistor input
18	Clarification to raise with the SwapCell project: the pack should move from legacy discharge (state 5) to heartbeat discharge (mode 2) without opening the output	Flag for SwapCell; not changed here. The interface is governed in the swapcell repo

■ Consequences

- SunSpoke builds to SwapCell interface v0.3 and cites items W, C and V. A new requirement R13 records this.
- R1 now states the 20 km/h default cut-off, the walk assist and the ban on a throttle. R12 is redefined; `budget_usd` stays at 250.
- SSP-PRB-001, SSP-PRC-001 and SSP-REQ-001 move to v0.3. TRL is set to 3. No TRL 4 work is started.